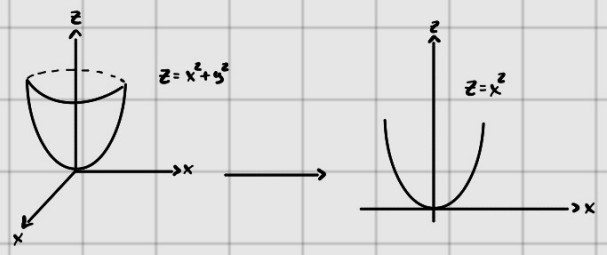
SUPERFICI DI ROTAZIONE

SE UNA SUPERFICIE È IL GRAFICO DI UNA FUNZIONE, OSSIA È DESCRITTA DA UN'EQUAZIONE $z = f(x,y)$ (PENSATE A $z = 8 - 3x + y$ O $z = x^2 - y^2$, ETC...) ALLORA SE $f(x,y)$ CONTIENE LE VARIABILI x E y SOTTO FORMA DEL POLINOMIO x^2+y^2, ALLORA LA SUPERFICIE È OTTENUTA RUOTANDO LA CURVA DEL PIANO ZX CHE HA EQUAZIONE UGUALE A $z = f(x,y)$, MA CON LA SOSTITUZIONE $x^2 \to x^2+y^2$

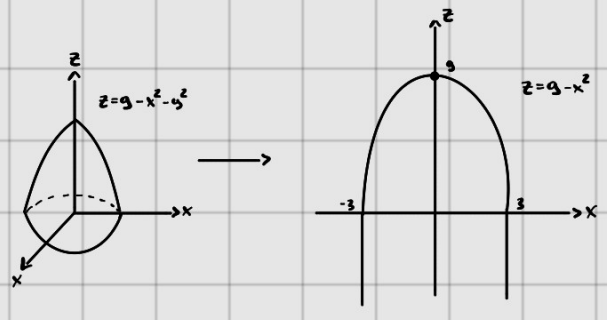
ESERCIZIO

1) $z = x^2+y^2$ È DI ROTAZIONE ED È OTTENUTA RUOTANDO $z = x^2$



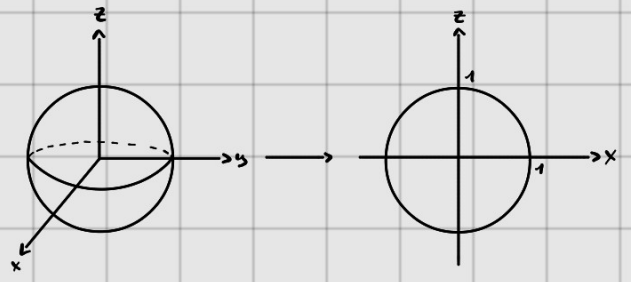
ESERCIZIO

2) $z = 9 - (x^2-y^2)$ È LA ROTAZIONE DELLA CURVA $z = 9 - x^2$



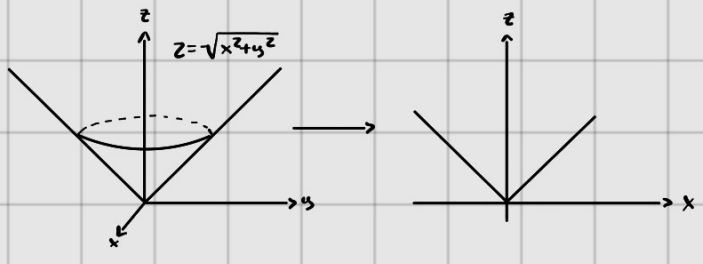
ESERCIZIO

3) $x^2+y^2+z^2 = 1$ È LA ROTAZIONE DI $x^2+z^2 = 1$



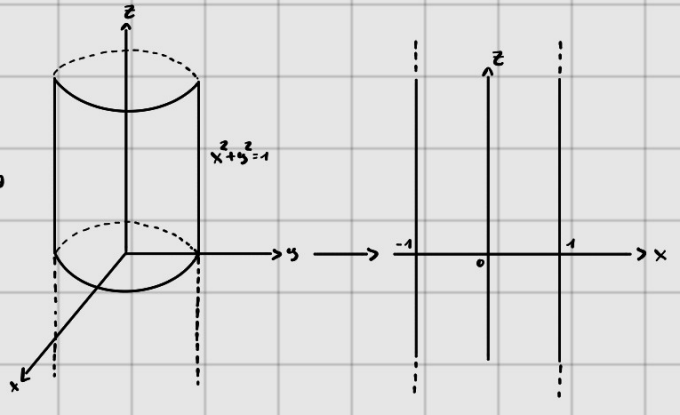
ESERCIZIO

4) $z = \sqrt{x^2+y^2}$ È OTTENUTO RUOTANDO $z = \sqrt{x^2} = |x|$



ESERCIZIO

5) $x^2+y^2 = 1$? È LA ROTAZIONE DI $x^2 = 1$, MA ESSA RAPPRESENTA LE 2 CURVE $x = +1$, $x = -1$



INTEGRALI SUPERFICIALI

Σ: SUPERFICIE REGOLARE PARAMETRIZZATA

DIFFERENZA TRA CURVE E SUPERFICI:

- PER LE CURVE CI INTERESSA IL VETTORE VELOCITÀ (TANGENTE)

$\int_\Sigma f(x,y,z)\,d\sigma$ $d\sigma$: ELEMENTO D'AREA

- PER LE SUPERFICI CALCOLEREMO IL VETTORE NORMALE

ESERCIZIO

$\Sigma = \left\{ z = 9 - x^2 - y^2,\ x^2 + y^2 \le \frac{1}{2} \right\}$ $\text{AREA}(\Sigma) = \int_\Sigma 1\,d\sigma$

CALCOLIAMO Σ, QUANDO Σ È IL GRAFICO DI UNA FUNZIONE CONVIENE PORRE $\Sigma \begin{cases} x = u \\ y = v \\ z = 9 - u^2 - v^2 \end{cases}$ $(u,v) \in \mathcal{D} = \left\{ u^2 + v^2 \le \frac{1}{2} \right\}$

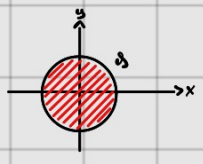
CALCOLIAMO ORA L'ELEMENTO D'AREA

$d\sigma = \|N\|\,du\,dv$ N = VETTORE NORMALE A Σ $N = \Sigma_u \wedge \Sigma_v$

$\Sigma_u = (1, 0, -2u) \longrightarrow \Sigma_u = \frac{\partial \Sigma}{\partial u}(u, v, 9 - u^2 - v^2)$

$\Sigma_v = (0, 1, -2v) \longrightarrow \Sigma_u = \frac{\partial \Sigma}{\partial v}(u, v, 9 - u^2 - v^2)$

$N = \Sigma_u \wedge \Sigma_v = \begin{vmatrix} i & j & k \\ 1 & 0 & -2u \\ 0 & 1 & -2v \end{vmatrix} \longrightarrow \text{SET}(N) = (2u, 2v, 1)$ $\|N\| = \sqrt{(2u)^2 + (2v)^2 + 1^2} = \sqrt{4u^2 + 4v^2 + 1} \Longrightarrow \int_\Sigma d\sigma = \iint_{\mathcal{D}} \sqrt{4u^2 + 4v^2 + 1}\,du\,dv$



COORDINATE POLARI $\begin{cases} u = \rho\cos\theta \\ v = \rho\operatorname{sen}\theta \end{cases}$ $\rho \in [0, \sqrt{\frac{1}{2}}]$ IL RAGGIO $\theta \in [0, 2\pi]$ SEMPRE QUESTO VALORE

$\text{Jac} = \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -\rho\operatorname{sen}\theta \\ \operatorname{sen}\theta & \rho\cos\theta \end{vmatrix}$

$\text{SET}(\text{Jac}) = (\cos\theta \cdot \rho\cos\theta) - (-\rho\operatorname{sen}\theta \cdot \operatorname{sen}\theta)$
$= \rho\cos^2\theta + \rho\operatorname{sen}^2\theta = \rho(\underbrace{\cos^2\theta + \operatorname{sen}^2\theta}_{1}) = \rho$

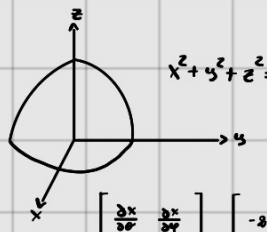
$\iint_{\mathcal{D}} \sqrt{4u^2 + 4v^2 + 1}\,du\,dv \longrightarrow \iint_{\mathcal{D}} \sqrt{4\rho^2\cos^2\theta + 4\rho^2\operatorname{sen}^2\theta + 1}\,\rho\,d\rho\,d\theta =$

$4\rho(\underbrace{\rho\cos^2\theta + \rho\operatorname{sen}^2\theta}_{1})$
$4\rho(\rho) = 4\rho^2 + 1$

$\iint_{\mathcal{D}} \sqrt{4\rho^2 + 1}\,d\rho\,d\theta = \int_0^{\sqrt{\frac{1}{2}}} d\rho \int_0^{2\pi} \rho\sqrt{4\rho^2 + 1}\,d\theta = 2\pi \int_0^{\sqrt{\frac{1}{2}}} \rho\sqrt{4\rho^2 + 1}\,d\rho = 2\pi \left[\frac{(4\rho^2 + 1)^{\frac{1}{2}}}{\frac{3}{4} \cdot 8} \right]_0^{\sqrt{\frac{1}{2}}} = \frac{2\pi}{6} \left[(4\rho^2 + 1)^{\frac{3}{2}} \right]_0^{\sqrt{\frac{1}{2}}} = \frac{\pi}{6}\left(3^{\frac{3}{2}} - 1\right) \longrightarrow \frac{\pi}{6}\left(\sqrt{3^3} - 1\right)$

ESERCIZIO

$\int_\Sigma \frac{d\sigma}{\sqrt{x^2 + y^2 + z^2}}$ IL PRIMO OTTANTE È DEFINITO DA $x \ge 0,\ y \ge 0,\ z \ge 0$



$x^2 + y^2 + z^2 = 4$

SFERA DI CENTRO $(0,0,0)$ E DI RAGGIO 2

COORDINATE SFERICHE $\Sigma : \begin{cases} x = 2\cos\theta\operatorname{sen}\varphi \\ y = 2\operatorname{sen}\theta\operatorname{sen}\varphi \\ z = 2\cos\varphi \end{cases}$ $\theta \in [0, \frac{\pi}{2}]$ $\varphi \in [0, \frac{\pi}{2}]$

$\text{Jac} = \begin{bmatrix} \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \varphi} \\ \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \varphi} \\ \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \varphi} \end{bmatrix} = \begin{bmatrix} -2\operatorname{sen}\theta\operatorname{sen}\varphi & 2\cos\theta\cos\varphi \\ 2\cos\theta\operatorname{sen}\varphi & 2\operatorname{sen}\theta\cos\varphi \\ 0 & -2\operatorname{sen}\varphi \end{bmatrix}$

$\|N\| = \|(-2\operatorname{sen}\theta\operatorname{sen}\varphi, 2\cos\theta\operatorname{sen}\varphi, 0) \wedge (2\cos\theta\cos\varphi, 2\operatorname{sen}\theta\cos\varphi, -2\operatorname{sen}\varphi)\| = \|(4\cos\theta\operatorname{sen}^2\varphi, -4\operatorname{sen}\theta\operatorname{sen}^2\varphi, -4\cos\varphi\operatorname{sen}\varphi)\| = 4\operatorname{sen}\varphi$

$\int_\Sigma \frac{d\sigma}{\sqrt{x^2 + y^2 + z^2}} = \int_\Sigma \sqrt{(2\cos^2\theta\operatorname{sen}^2\varphi + 2\operatorname{sen}^2\theta\operatorname{sen}^2\varphi + 2\cos^2\varphi)}$

$\sqrt{2\operatorname{sen}^2\varphi(\underbrace{\cos^2\theta + \operatorname{sen}^2\theta}_{1}) + 2\cos^2\varphi}$

$\sqrt{4(\operatorname{sen}^2\varphi + \cos^2\varphi)} = \sqrt{4}$

$\|N\| = \|(-2\sin\sigma\sin\varphi, 2\cos\sigma\sin\varphi, 0) \wedge (2\cos\sigma\cos\varphi, 2\sin\sigma\cos\varphi, -2\sin\varphi)\|$

$= \|(4\cos\sigma\sin^2\varphi, -4\sin\sigma\sin^2\varphi, -4\cos\varphi\sin\varphi)\| = 4\sin\varphi$

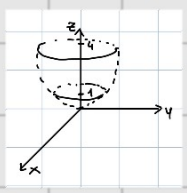
$\int_\Sigma \frac{d\sigma}{\sqrt{x^2 + y^2 + z^2}} = \iint_D \frac{1}{\sqrt{4}} \cdot 4\sin\varphi\,d\sigma\,d\varphi$ $D = \left[0, \frac{\pi}{2}\right] \times \left[0, \frac{\pi}{2}\right]$

$= \iint_{\mathcal{D}} \frac{1}{\sqrt{4}} 4\sin\varphi\,d\sigma\,d\varphi = 2\int_0^{\frac{\pi}{2}} d\sigma \int_0^{\frac{\pi}{2}} \sin\varphi\,d\varphi = 2\left[\sigma\right]_0^{\frac{\pi}{2}} \cdot \left[-\cos\varphi\right]_0^{\frac{\pi}{2}} = 2\left(\frac{\pi}{2} - 0\right) \cdot (-0 + 1) =$

$= \boxed{\pi}$

ESERCIZIO

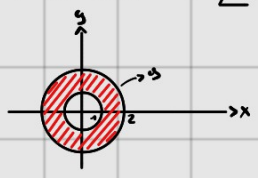
$$\int_\Sigma \operatorname{arctg}\frac{y}{x}\,d\sigma = \iint_{\mathcal{D}} \left(\operatorname{arctg}\frac{v}{u}\right)\sqrt{1+4u^2+4v^2}\,du\,dv \qquad \Sigma = \left\{ z = x^2+y^2,\ 1 \le z \le 4 \right\}$$



$z = x^2+y^2$ È DI ROTAZIONE OTTENUTA DALLA CURVA $z = x^2$



$$\Sigma : \begin{cases} x = u \\ y = v \\ z = u^2+v^2 \end{cases} \qquad (u,v) \in \mathcal{D} = ?$$



$\mathcal{D}$ LO OTTENIAMO DALLA RELAZIONE $1 \le z \le 4$ SOSTITUENDO (z) CON $z = x^2+y^2 \Rightarrow 1 \le x^2+y^2 \le 4$ $\quad x^2+y^2 = 1$ $\quad x^2+y^2 = 4$

IN CASI (COME QUESTO) DOVE Σ È IL GRAFICO DI UNA FUNZIONE $(z = f(x,y))$ $\quad d\sigma = \sqrt{1+f_x^2+f_y^2}\,du\,dv$

$d\sigma = \|N\|\,du\,dv = \|\Sigma_u \wedge \Sigma_v\|\,du\,dv$. NEL NOSTRO CASO $z = x^2+y^2 \Rightarrow f(x,y) = x^2+y^2$ $\quad f_x = 2x$ $\quad f_y = 2y$

$$d\sigma = \sqrt{1+f_x^2+f_y^2}\,du\,dv = \sqrt{1+(2u)^2+(2v)^2}\,du\,dv = \sqrt{1+4u^2+4v^2}\,du\,dv$$

$$\int_\Sigma \operatorname{arctg}\frac{y}{x}\,d\sigma = \iint_{\mathcal{D}} \left(\operatorname{arctg}\frac{v}{u}\right)\sqrt{1+4u^2+4v^2}\,du\,dv \xrightarrow{\text{COORDINATE POLARI}} \begin{cases} u = \rho\cos\theta & \rho = [1,2] \\ v = \rho\operatorname{sen}\theta & \theta = [0,2\pi] \end{cases} \qquad \text{Jac} = \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -\rho\operatorname{sen}\theta \\ \operatorname{sen}\theta & \rho\cos\theta \end{vmatrix}$$

$$\iint_{\mathcal{D}} \left(\operatorname{arctg}\frac{\rho\operatorname{sen}\theta}{\rho\cos\theta}\right)\sqrt{1+4\rho^2\underbrace{\cos^2\theta}_{1}+4\rho^2\underbrace{\operatorname{sen}^2\theta}_{}}\,\rho\,d\rho\,d\theta = \iint_{\mathcal{D}} \left(\operatorname{arctg}\frac{\operatorname{sen}\theta}{\cos\theta}\right)\sqrt{1+4\rho^2}\,\rho\,d\rho\,d\theta = \iint \theta\cdot\rho\sqrt{1+4\rho^2}\,d\rho\,d\theta = \int_0^{2\pi}\theta\,d\theta\int_1^2 \rho\sqrt{1+4\rho^2}\,d\rho =$$

$\operatorname{tg}\theta$



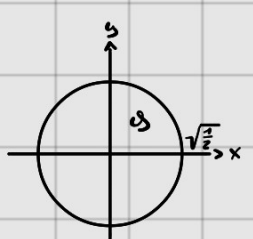
$$= \int_0^{2\pi}\theta\left[\frac{(1+4\rho^2)^{\frac{1}{2}+1}}{\frac{3}{2}\cdot 8}\right]_1^2 d\theta = \left[\frac{(1+4\rho^2)^{\frac{3}{2}}}{12}\right]\int_0^{2\pi}\theta\,d\theta = \left(\frac{17^{\frac{3}{2}}}{12}-\frac{5^{\frac{3}{2}}}{12}\right)\cdot\left[\frac{\theta^2}{2}\right]_0^{2\pi} = \frac{1}{12}\left(17^{\frac{3}{2}}-5^{\frac{3}{2}}\right)\cdot\left(\frac{4\pi^2}{2}-0\right) = \frac{\pi^2}{6}\left(\sqrt{7^3}-\sqrt{5^3}\right)$$

$\rho\sqrt{1+4\rho^2} = \rho(1+4\rho^2)^{\frac{1}{2}}$

ESERCIZIO

CALCOLARE L'AREA DELLA PORZIONE DI SUPERFICIE $\Sigma = \left\{ z = 9-x^2-y^2,\ x^2+y^2 \le \frac{1}{2} \right\}$

$$A = \iint_{\mathcal{D}_{xy}} \sqrt{1+(f'_x)^2+(f'_y)^2}\,dx\,dy \Rightarrow \iint_{\mathcal{D}} \sqrt{1+(-2x)^2+(-2y)^2}\,dx\,dy \qquad \mathcal{D}: \begin{cases} x = \rho\cos\theta & 0 \le \rho \le \sqrt{\frac{1}{2}} \\ y = \rho\operatorname{sen}\theta & 0 \le \theta \le 2\pi \end{cases}$$

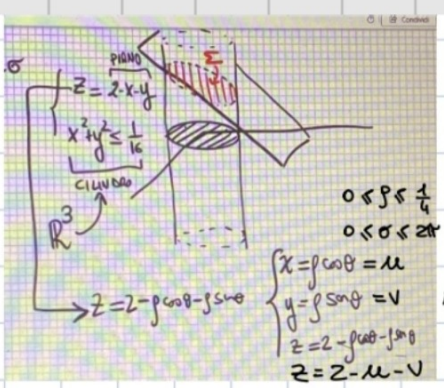


DET DELLA MATRICE

$$\int_0^{2\pi} d\theta \int_0^{\sqrt{\frac{1}{2}}} \sqrt{1+4\rho^2\underbrace{\cos^2\theta}_{1}+4\rho^2\operatorname{sen}^2\theta}\;\rho\,d\rho = \int_0^{2\pi} d\theta \int_0^{\sqrt{\frac{1}{2}}} \sqrt{1+4\rho^2+4\rho^2}\,d\rho = \int_0^{2\pi} d\theta \int_0^{\sqrt{\frac{1}{2}}} \sqrt{1+4\rho^2}\,\rho\,d\rho = \frac{1}{8}\int_0^{2\pi} d\theta \int_0^{\sqrt{\frac{1}{2}}} \underbrace{8\rho}_{f'}\,\underbrace{(1+4\rho^2)^{\frac{1}{2}}}_{f}\,d\rho = \frac{1}{8}\int_0^{2\pi} d\theta \left[\frac{(1+4\rho^2)^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^{\sqrt{\frac{1}{2}}} =$$

$$= \frac{1}{8}\int_0^{2\pi}\frac{2}{3}\sqrt{27}-\frac{2}{3}\,d\theta = \frac{1}{8}\left[\frac{2}{3}\sqrt{27}\,d\theta-\frac{2}{3}\theta\right]_0^{2\pi} = \frac{1}{8}\left(\frac{2}{3}\sqrt{27}\cdot 2\pi-\frac{2}{3}\cdot 2\pi\right) = \frac{\sqrt{27}\,\pi}{6}-\frac{\pi}{6}$$

ESERCIZIO



CALCOLARE L'INTEGRALE SUPERFICIALE $\int_\Sigma (x+2y+z-4)\,d\sigma$

DOVE $\Sigma = \left\{ z = 2-x-y,\ x^2+y^2 \le \frac{1}{16} \right\}$

$x^2+y^2 \le \frac{1}{4}$ $\quad r = \frac{1}{4}$

$$(u,v) \in \mathcal{D} = \left\{ u^2+v^2 \le \frac{1}{16} \right\}$$

$\Sigma_u = (1,0,-1)$ $\qquad \Sigma_v = (0,1,-1)$

$$N = \Sigma_u \wedge \Sigma_v = \begin{vmatrix} i & j & k \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix} = (1,-1,1)$$

$$\|N\| = \sqrt{1^2+(-1)^2+1^2} = \sqrt{3}$$

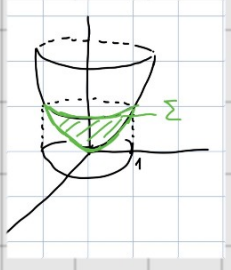
$$\int_\Sigma (x+2y+z-4)\,d\sigma = \iint_D (u+2v+2-u-v-4)\cdot\sqrt{3}\;du\,dv = \iint_D (v-2)\cdot\sqrt{3}\;du\,dv =$$

$$-2\sqrt{3}\int_0^{2\pi} du \int_0^{\frac{1}{4}} v\,dv = -2\sqrt{3}\int_0^{2\pi}\left[\frac{v^2}{2}\right]_0^{\frac{1}{4}} du = -2\sqrt{3}\int_0^{2\pi}\frac{1}{16}\cdot\frac{1}{2}\,du = -2\sqrt{3}\int_0^{2\pi}\frac{1}{32}\,du =$$

$$-2\sqrt{3}\left[\frac{1}{32}u\right]_0^{2\pi} = -2\sqrt{3}\cdot\frac{1}{32}\cdot 2\pi = \boxed{-\frac{\sqrt{3}\pi}{8}}$$

ESERCIZIO

CALCOLARE L'INTEGRALE SUPERFICIALE $\int_\Sigma \frac{1}{1+4z}$ DOVE $\Sigma = \{z = x^2+y^2,\ x^2+y^2 \le 1\}$



$$\int_\Sigma \frac{1}{1+4z} = \iint_D \frac{1}{1+4(x^2+y^2)}\cdot\sqrt{1+4x^2+4y^2}\,dx\,dy$$

$d\sigma = \sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}\,dx\,dy$

$d\sigma = \sqrt{1+4\rho^2\cos^2\theta+4\rho^2\sin^2\theta}\,\rho\,d\theta\,d\rho$

$d\sigma = \sqrt{1+4\rho^2}\,\rho\,d\rho\,d\theta$

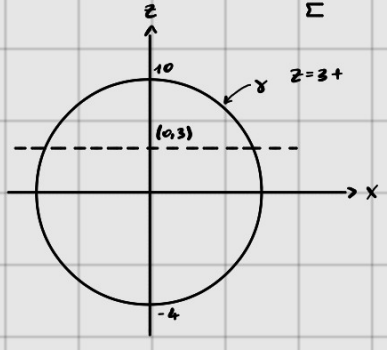
PARAMETRIZZO $\begin{cases} x = \rho\cos\theta & \rho\in[0,1] \\ y = \rho\sin\theta & \theta\in[0,2\pi] \end{cases}$

$$\int_0^{2\pi} d\theta \int_0^1 \frac{1}{1+4(\rho^2\cos^2\theta+\rho^2\sin^2\theta)}\cdot\sqrt{1+4\rho^2\cos^2\theta+4\rho^2\sin^2\theta}\;\rho\,d\rho = \int_0^{2\pi} d\theta\int_0^1 \frac{1}{1+4\rho^2}\cdot\sqrt{1+4\rho^2}\;\rho\,d\rho = \int_0^{2\pi} d\theta\int_0^1 \frac{\sqrt{1+4\rho^2}}{\sqrt{1+4\rho^2}\cdot\sqrt{1+4\rho^2}}\;\rho\,d\rho = \int_0^{2\pi} d\theta\int_0^1 \frac{\rho}{\sqrt{1+4\rho^2}}\,d\rho =$$

$$= \frac{1}{8}\int_0^{2\pi} d\theta\int_0^1 8\rho(1+4\rho^2)^{-\frac{1}{2}} = \frac{1}{8}\int_0^{2\pi} d\theta\left[\frac{(1+4\rho^2)^{\frac{1}{2}}}{\frac{1}{2}}\right]_0^1 = \frac{1}{8}(2\sqrt{5}-2)\int_0^{2\pi} d\theta = \frac{1}{8}(2\sqrt{5}-2)\,2\pi = \frac{\pi}{4}(2\sqrt{5}-2)$$

ESERCIZIO

CALCOLARE L'INTEGRALE $\int_\Sigma xy(z-3)\,d\sigma$ DOVE Σ È LA PORZIONE DI CONO $z = 3+\sqrt{49-x^2-y^2}$ INTERNA AL CILINDRO $x^2+y^2=4$



Σ È LA ROTAZIONE DI γ

$z = 3+\sqrt{49-x^2}$

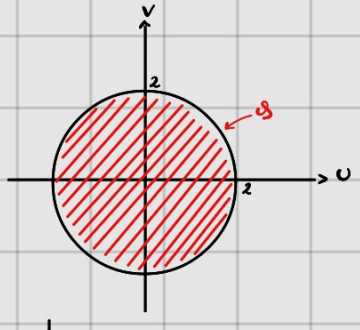
$z-3 = \sqrt{49-x^2}$

$(z-3)^2 = 49-x^2$

$x^2+(z-3)^2 = 49$

PARAMETRIZZO $\begin{cases} x = u \\ y = v \\ z = 3+\sqrt{49-u^2-v^2} \end{cases}$

$(u,v)\in D = \{u^2+v^2\le 4\}$



$$\Sigma_u = \left(1,0,\frac{-u}{\sqrt{49-u^2-v^2}}\right) \longrightarrow \Sigma_u = \frac{\partial\Sigma}{\partial u}\left(u,v,3+\sqrt{49-u^2-v^2}\right)$$

$$\Sigma_v = \left(0,1,\frac{-v}{\sqrt{49-u^2-v^2}}\right) \longrightarrow \Sigma_v = \frac{\partial\Sigma}{\partial v}\left(u,v,3+\sqrt{49-u^2-v^2}\right)$$

$$N = \Sigma_u\wedge\Sigma_v = \begin{vmatrix} i & j & k \\ 1 & 0 & \frac{-u}{\sqrt{49-u^2-v^2}} \\ 0 & 1 & \frac{-v}{\sqrt{49-u^2-v^2}} \end{vmatrix} = \left(\frac{u}{\sqrt{49-u^2-v^2}},\frac{-v}{\sqrt{40-u^2-v^2}},1\right)$$

$$\|N\| = \sqrt{\left(\frac{-u}{\sqrt{49-u^2-v^2}}\right)^2+\left(\frac{-v}{\sqrt{49-u^2-v^2}}\right)^2+1^2} = \sqrt{\frac{u^2+v^2+49-u^2-v^2}{49-u^2-v^2}} = \sqrt{\frac{49}{49-u^2-v^2}} = \frac{7}{\sqrt{49-u^2-v^2}}$$

PARAMETRIZZO $\begin{cases} x = \rho\cos\theta & \theta\in[0,2\pi] \\ y = \rho\sin\theta & \rho\in[0,2] \end{cases}$ $du\,dv = \rho\,d\rho\,d\theta$

$\det = \rho$

$$\int_\Sigma xy(z-3)\,d\sigma = \iint_D uv\left(3+\sqrt{49-u^2-v^2}-3\right)\cdot\frac{7}{\sqrt{49-u^2-v^2}}\,du\,dv = \iint_D 7uv\,du\,dv = 7\iint_D uv\,du\,du = 7\iint_D \rho\cos\theta\,\rho\sin\theta\,\rho\,d\theta\,d\rho =$$

$$= 7\int_0^{2\pi} d\theta\int_0^2 \rho\cos\theta\,\rho\sin\theta\,\rho\,d\rho = 7\int_0^{2\pi}\cos\theta\sin\theta\,d\theta\int_0^2 \rho^3\,d\rho = 7\int_0^{2\pi}\cos\theta\sin\theta\,d\theta\cdot\left[\frac{\rho^4}{4}\right]_0^2 = \frac{16}{4}\cdot 7\left[\frac{\sin^2\theta}{2}\right]_0^{2\pi} = 28(0-0) = 0$$

ESERCIZIO

DATA $\Sigma = \{ z = x - 2y,\ x^2+y^2 \le 4 \}$ CALCOLARE $\int_\Sigma x z^2 \, d\sigma$

$\int_\Sigma x z^2 \, d\sigma \qquad z = x - 2y \qquad x^2+y^2 = 4$

$$\iint_A \sqrt{1 + (z_x')^2 + (z_y')^2}\ (x z^2)\, dx dy = \iint_A \sqrt{1+1+(-2)^2} \cdot \left[x (x-2y)^2 \right] dx dy = \iint_A \sqrt{6} \left[x (x^2 + 4y^2 - 4xy) \right] dx dy = \iint_A \sqrt{6} \cdot (x^3 + 4xy^2 - 4x^2 y)\, dx dy$$

PARAMETRIZZO

$\Phi : \begin{cases} x = \rho\cos\theta & \rho \in [0,2] \\ y = \rho\sin\theta & \theta \in [0, 2\pi] \end{cases}$ $\quad x^2+y^2 \le 4 \longrightarrow x^2+y^2 \le \sqrt{4}$ $\quad x^2+y^2 \le r^2$

$$\int_0^{2\pi} \cos(1 - \operatorname{sen}^2\theta)\, d\theta \longrightarrow \left[-\int -\operatorname{sen}(2\theta)\operatorname{sen}(\theta)\, d\theta + \cos^2\theta \cdot \operatorname{sen}\theta \right.$$

DET JAC

$$\int_0^{2\pi} d\theta \int_0^2 \sqrt{6} \left(\rho^3 \cos^3\theta + 4\rho\cos\theta \cdot \rho^2 \operatorname{sen}^2\theta - 4\rho^2 \cos^2\theta \cdot \rho\operatorname{sen}\theta \right) \rho\, d\rho = \sqrt{6} \int_0^{2\pi} d\theta \int_0^2 \rho^4 \cos^3\theta + 4\rho^4 \cos\theta \cdot \operatorname{sen}^2\theta - 4\rho^4 \cos^2\theta \cdot \operatorname{sen}\theta\, d\rho =$$

$$= \sqrt{6} \int_0^{2\pi} \cos^3\theta + 4\cos\theta \operatorname{sen}^2\theta - 4\cos^2\theta \operatorname{sen}\theta\, d\theta \int_0^2 \rho^4\, d\rho = \sqrt{6} \int_0^{2\pi} \cos^3\theta + \int_0^{2\pi} 4\cos\cdot\operatorname{sen}^2\theta - \int_0^{2\pi} 4\cos^2\theta \cdot \operatorname{sen}\theta\, d\theta \cdot \left[\frac{\rho^5}{5} \right]_0^2 =$$

$$= \sqrt{6} \cdot \frac{32}{5} \int_0^{2\pi} \cos\cdot\cos^2\theta + 4\cos\cdot\operatorname{sen}^2\theta - 4\cos^2\theta \cdot \operatorname{sen}\theta\, d\theta = \sqrt{6} \cdot \frac{32}{5} \int_0^{2\pi} \cos(1 - \operatorname{sen}^2\theta)\, d\theta + \left[4 \cdot \frac{\operatorname{sen}^3\theta}{3} + 4 \cdot \frac{\cos^3\theta}{3} \right]_0^{2\pi} =$$

$$= \sqrt{6} \cdot \frac{32}{5} \left[-\left(-\frac{2}{3}\operatorname{sen}^3\theta \right) + 3\cos^2\theta \cdot \operatorname{sen}\theta + 4 \cdot \frac{\operatorname{sen}^3\theta}{3} + 4 \cdot \frac{\cos^3\theta}{3} \right]_0^{2\pi} = \sqrt{6} \cdot \frac{32}{5} \left[-\left(-\frac{2}{3} \cdot 0 \right) + 3 \cdot 1 \cdot 0 + 4 \cdot 0 + 4 \cdot \frac{1}{3} \right] - \left[-\left(-\frac{2}{3} \cdot 0 \right) + 3 \cdot 1 \cdot 0 + 4 \cdot 0 + 4 \cdot \frac{1}{3} \right] =$$

$$= \sqrt{6} \cdot \frac{32}{5} \left(\frac{4}{3} - \frac{4}{3} \right) = 0$$